Starter

Find the general solution to:

1. $$\frac{\mathrm{d}y}{\mathrm{d}x} + 2y = x$$

2. $$\ddot{x} + \dot{x} - 6x = 0$$

3. Find the particular solution to

$$\frac{\mathrm{d}^2 z}{\mathrm{d}t^2} - 9z = 0, \text{given } \left.\frac{\mathrm{d}z}{\mathrm{d}t}\right|_{t=0} = 16, z|_{t=0} = 4$$

Solutions

1. $\mathrm{IF} = e^{2x}, \therefore e^{2x}\dfrac{\mathrm{d}y}{\mathrm{d}x} + 2e^{2x}y = xe^{2x}$

$$e^{2x}y = \int xe^{2x}\mathrm{d}x = \frac{1}{2}xe^{2x} - \int \frac{1}{2}e^{2x}\mathrm{d}x$$

$$= \frac{1}{2}xe^{2x} - \frac{1}{4}e^{2x} + c$$

$$\therefore y = \frac{1}{2}x - \frac{1}{4} + ce^{-2x}$$

Solutions

2. $\mathrm{AE} = m^2 + m - 6 = 0, m \in \{2, -3\},$

$$\therefore x = Ae^{2t} + Be^{-3t}$$

Solutions

3. $$\mathrm{AE} = m^2 - 9 = 0, \quad m = 3, \quad \therefore x = Ae^{3t} + Bte^{3t}$$

$$\dot{x} = 3Ae^{3t} + Be^{3t} + 3Bte^{3t}$$

$$x|_{t=0} = 4 = A + 0 \Rightarrow A = 4$$

$$\dot{x}|_{t=0} = 12 = 3A + B + 0 \Rightarrow B = 4$$

$$\therefore x = 4e^{3t} + 4te^{3t}$$

Differential Equations 2

2nd Order Homogeneous 2

Purely Imaginary Roots

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$$\begin{aligned}
y &= Ae^{3\mathrm{i}x} + Be^{-3\mathrm{i}x} \\
&= A(\cos 3x + \mathrm{i}\sin 3x) + B(\cos -3x + \mathrm{i}\sin -3x) \\
&= A(\cos 3x + \mathrm{i}\sin 3x) + B(\cos 3x - \mathrm{i}\sin 3x) \\
&= (A + B)\cos 3x + (A - B)\ \mathrm{i}\sin 3x \\
&= C\cos 3x + D\sin 3x
\end{aligned}$$

Purely Imaginary roots in General

Show that the general solution to

$$\frac{\mathrm{d}^2 y}{\mathrm{d}x^2} + \omega^2 y = 0$$

can be written in the form

$$y = A\cos\omega x + B\ \mathrm{i}\sin\omega x$$

where A, B, ω are real numbers.

Complex roots

Can we extend this to a complex root, $\lambda \pm \mu \, \mathrm{i}$?

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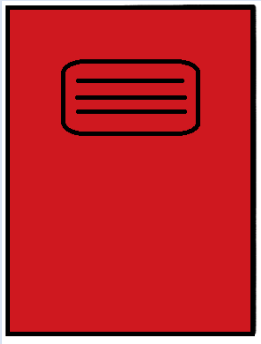
$$
\begin{aligned}
y &= Ae^{(\lambda+\mu\ \mathrm{i})x} + Be^{(\lambda-\mu\ \mathrm{i})x} \\
&= Ae^{\lambda x}e^{\mu \mathrm{i} x} + Be^{\lambda x}e^{-\mu \mathrm{i} x} \\
&= e^{\lambda x}\left(Ae^{\mu \mathrm{i} x} + Be^{-\mu \mathrm{i} x}\right) \\
&= e^{\lambda x}(C\cos\mu x + D\sin\mu x)
\end{aligned}
$$

Sketching the solutions



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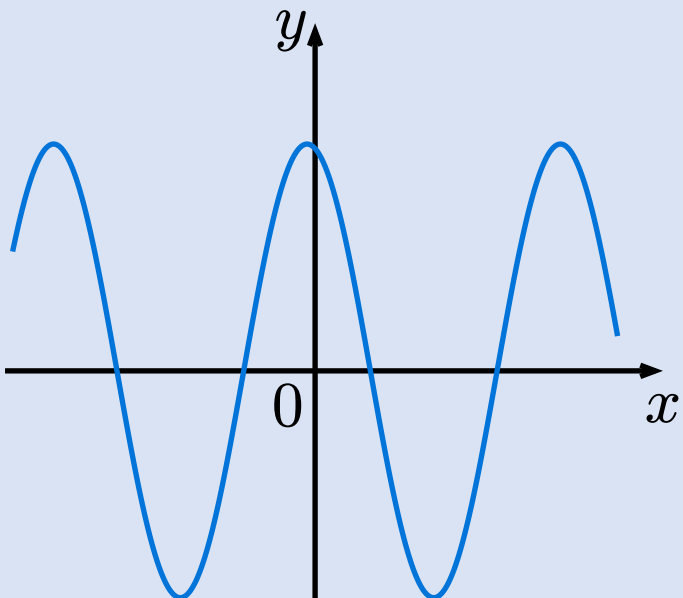
$$y = A\cos\mu x + B\sin\mu x \equiv R\cos(x+\alpha)$$



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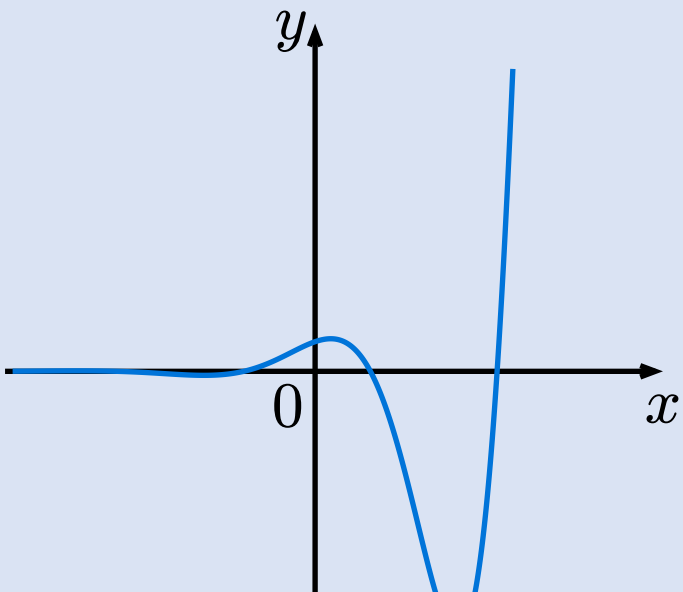

Finding a Particular Solution

Find and sketch the particular solution to

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$$\text{AE: } m^2 - 2m + 2 = 0 \Rightarrow m = 1 \pm \mathrm{i}$$

$$\therefore x = e^t(A\cos t + B\sin t)$$

$$\therefore \dot{x} = e^t(A\cos t + B\sin t) + e^t(B\cos t - A\sin t)$$

$$x|_{t=0} = 1 = e^0(A\cos 0 + B\sin 0) \Rightarrow A = 1$$

$$\dot{x}|_{t=0} = 2 = e^0(A\cos 0 + B\sin 0) + e^0(B\cos 0 - A\sin 0) \Rightarrow B = 1$$

$$\therefore x = e^t(\cos t + \sin t) = \sqrt{2}e^t\cos\left(t + \frac{\pi}{4}\right)$$

Finding a Particular Solution

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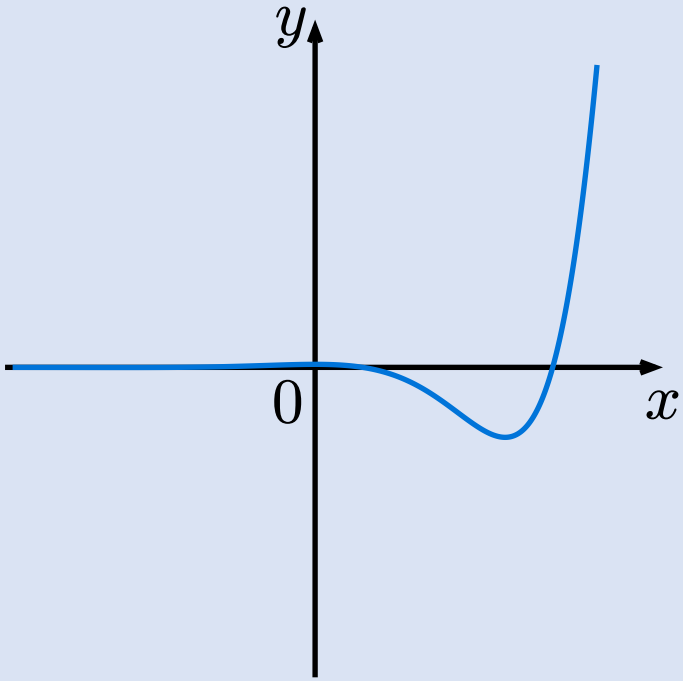


Recall

Given the equation

$$a\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + b\frac{\mathrm{d}y}{\mathrm{d}x} + cy = 0$$

state the four possible forms the general solution may take, and the conditions under which it will take them.

General Solutions

If the auxiliary equation, $am^2 + bm + c = 0$ have:

- Two distinct real roots, λ, μ:
  $y = Ae^{\lambda x} + Be^{\mu x}$
- Repeated real roots, λ:
  $y = Ae^{\lambda x} + Bxe^{\lambda x}$
- Purely imaginary roots, $\pm\omega\ \mathrm{i}$:
  $y = A\cos\omega x + B\sin\omega x \text{ or } R\cos(\omega x + \alpha)$
- Complex roots, $\alpha \pm \beta\ \mathrm{i}$:
  $y = e^{\alpha x}(A\cos\beta x + B\sin\beta x)$

Practice! (4s.f. where appropriate)

1. Find the general solution to the equation $\frac{d^2x}{dt^2} + 18\frac{dx}{dt} + 81x = 0$.
2. Find the general solution to the equation $\frac{d^2y}{dx^2} + 20\frac{dy}{dx} + 100y = 0$.
3. Find the general solution to the equation $5\ddot{x} + 3x = 0$.
4. Find the general solution to the equation $\ddot{x} + 18\dot{x} + 81x = 0$.
5. Find the general solution to the equation $5\frac{d^2p}{dq^2} + 3\frac{dp}{dq} + 10p = 0$.
6. Find the general solution to the equation $\frac{d^2y}{dx^2} + 6\frac{dy}{dx} + 8y = 0$.

Practice! (4s.f. where appropriate)

1. F
2. F
3. F
4. F
5. F
6. F

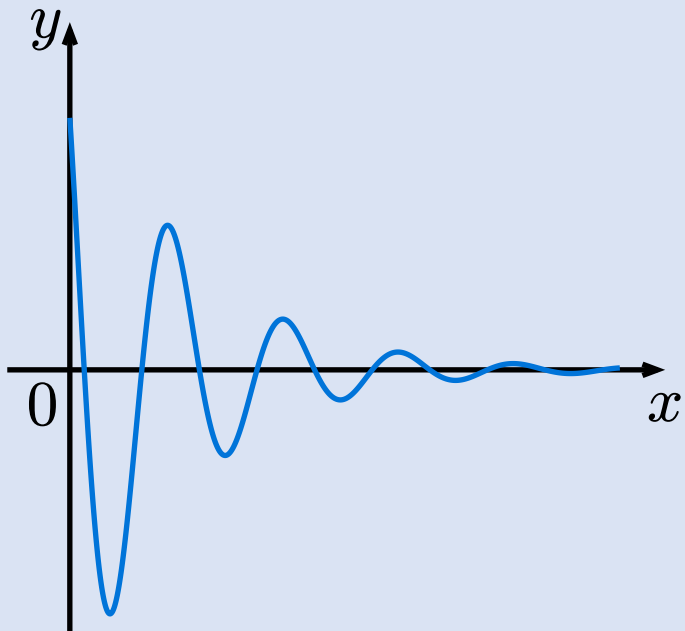
1. $x = Ae^{-9t} + Bte^{-9t}$
2. $y = Ae^{-10x} + Bxe^{-10x}$
3. $x = A\sin(0.7746t) + B\cos(0.7746t)$
4. $x = Ae^{-9t} + Bte^{-9t}$
5. $p = e^{-\frac{3}{10}q}(A\sin(1.382q) + B\cos(1.382q))$
6. $y = Ae^{-2x} + Be^{-4x}$

Damping

These second order DEs often describe real-world events. If you think about a spring, for instance, its acceleration is based on the distance from its resting position, but so is its velocity.

Many such systems attempt to reach an equilibrium position.

EG:



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Many suc on.

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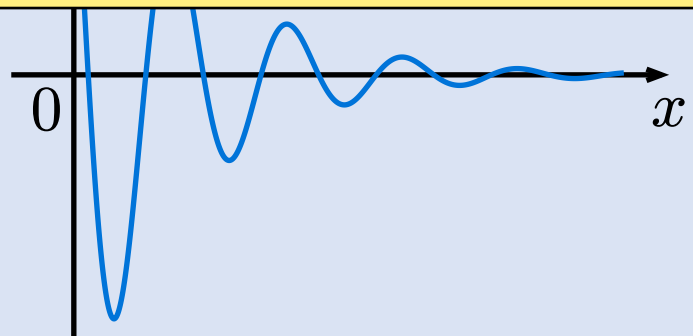
Moving Examples! 😀



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When discussing these solutions, we consider **under-damped**, **over-damped** and **critically damped** systems.

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The DEs all have particular solutions when these occur:

- **under-damped**: auxiliary equation has complex roots.
- **over-damped**: auxiliary equation has distinct real roots.
- **critically damped**: auxiliary equation has repeated negative roots.
- **undamped** systems are when the AE roots are purely imaginary (Simple Harmonic Motion).